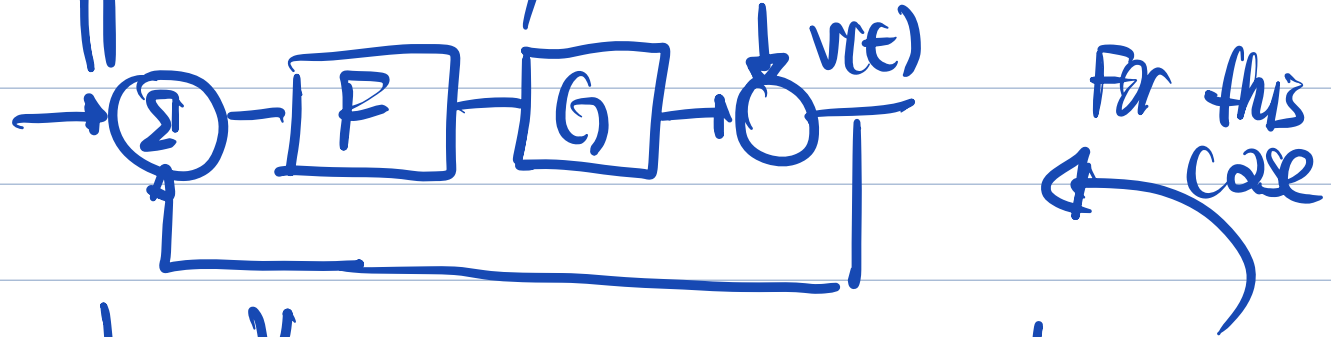


Quick Review

① SENSITIVITY

Feed back reduce the effect caused by an external disturbance $v(t)$



$$Y = \frac{F \cdot G}{1 + FG} \cdot R + \frac{1}{1 + FG} v$$

we call it
sensitivity function!

$$\Rightarrow S(s) = \frac{1}{1 + F(s)G(s)}$$

If $|S(s)| \ll 1 \Rightarrow$ little impact!

② ROBUSTNESS

Real systems behaves like

$$G^0(s) = G(s) [1 + \Delta G(s)]$$

error between real
and modeled/nominal

Under what condition the REAL MODEL is stable if the Nominal Model $G(s)$ is already stable?

ROBUSTNESS CRITERION

$$\left| \frac{F \cdot G(j\omega)}{1 + F \cdot G(j\omega)} \right| < \frac{1}{|\Delta G(j\omega)|}$$

Look at the ω which makes
the magnitude of the
 $|G_{cl}(j\omega)|$ high!!
(Bode peak)

③ STATE SPACE REPRESENTATION

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) + Du(t)\end{aligned}$$

(★)

where $x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ \vdots \\ x_n(t) \end{bmatrix}$

$u(t)$: inputs

$x(t)$ are internal VARIABLES which derivatives depends only in them ($x(t)$) and in the input $u(t)$

If we apply the Laplace transform in $\star$
we get

$$Y(s) = G(s) \cdot U(s) \Rightarrow G(s) = C(sI - A)^{-1} \cdot B + D$$

$$\therefore G(s) = \frac{C \operatorname{adj}(sI - A) B}{\det(sI - A)} + D$$

The eigenvalues of A gives the poles of the system.

6.1)

a) Sensitivity function

$$\rightarrow V - Y \cdot K \cdot G = Y \Rightarrow V = (1 + KG) \cdot Y$$

$$\Rightarrow \frac{Y}{V} = \frac{1}{(1 + KG)} = \frac{1}{\left(1 + K \frac{1}{s(s+1)}\right)}$$

$$= \frac{s(s+1)}{s^2 + s + K} \xrightarrow{s=j\omega} \frac{j\omega(j\omega+1)}{-\omega^2 + j\omega + K} = S(j\omega)$$

We compute the magnitude of $S(j\omega)$.

$$\Rightarrow |S(j\omega)| = \frac{\omega \sqrt{\omega^2 + 1}}{\sqrt{(K - \omega^2)^2 + \omega^2}}$$

For $\omega = 1$

b) Amplitude of
 $|V| = 1$

$$|S(j)| = \frac{\sqrt{2}}{\sqrt{(k-1)^2 + 1}}$$

we need $|y| < |V| = 1 \Rightarrow |y| = |S||V| < 1$

$$|y| = |T||V| \Rightarrow |y| < 1$$

$$\frac{\sqrt{2}}{\sqrt{(k-1)^2 + 1}} < 1$$

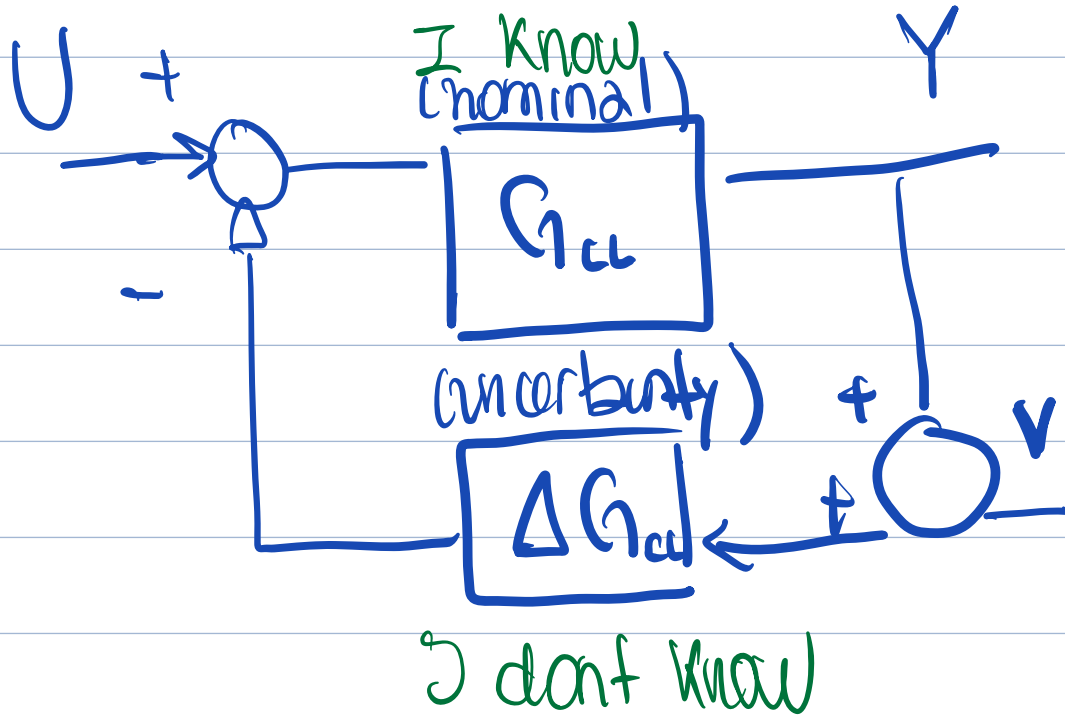
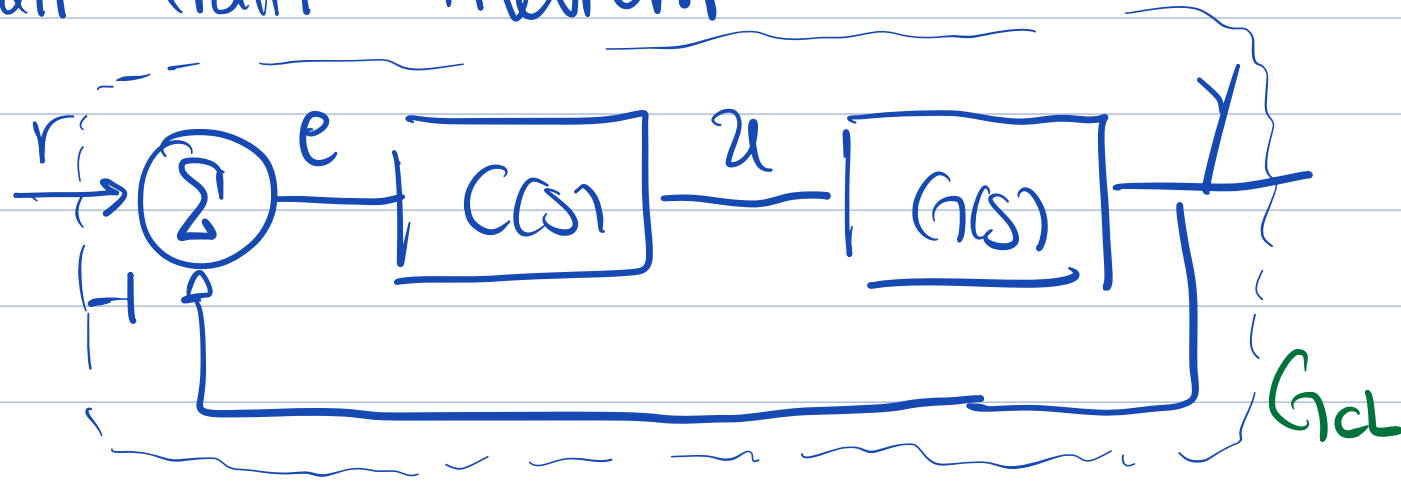
$$2 < (k-1)^2 + 1$$

$$1 < (k-1)^2$$

$$1 < k-1 \Rightarrow$$

$$k > 2$$

Small Gain theorem



$$\frac{Y}{U} = \frac{G_d}{(1 + G_d \cdot \Delta)}$$

$$\|G_d(i\omega) \cdot \Delta G(i\omega)\| < 1 \quad \forall \omega$$

$$\left\| \frac{C(j\omega)G(j\omega)}{1 + C(j\omega)G(j\omega)} \right\| = \|G_d(i\omega)\| < \frac{1}{\|\Delta G(i\omega)\|}$$

Relation between real plant vs nominal

$$\Delta G = \frac{G^0 - G_c}{G_c} \Rightarrow G_c \Delta G + G_c = G^0$$

$$G^0 = G_c (1 + \Delta G)$$

To draw root locus we need:

a) $1 + K \cdot G = 0$

$$1 + 4 \cdot \frac{1}{s(s+1)} \cdot \frac{\alpha}{(s+\alpha)} = 0$$

$$s^3 + \alpha s^2 + s^2 + \alpha s + 4\alpha = 0 \Rightarrow s^3 + s^2 + \alpha(s^2 + s + 4) = 0$$

$$1 + \alpha \frac{s^2 + s + 4}{s^3 + s^2} = 1 + \alpha \frac{s^2 + s + 4}{s^2(s+1)}$$

$K=\infty \Rightarrow$ $Q(s) = s^2 + s + 4 \Rightarrow$ ^{zeros} $-0.5 \pm j 1.94$ $\left. \begin{array}{l} m=2 \\ n=3 \end{array} \right\}$

$K=0 \Rightarrow P(s) = s^2(s+1) \Rightarrow s=0,0 \quad s=-1$

Locus max $(2,3) = 3 //$

Asymptotes $\downarrow$

$$\textcircled{2} \text{ Centroid: } \frac{\sum p_i - \sum z_i}{n-m} = -1 + 0.5 + 0.5 = 0$$

NOT NECESSARY TO CALCULATE SINCE $m=n$:

$$\textcircled{3} \angle A = \pi$$

$\textcircled{4}$ Cross in the Im axis ($s = j\omega$)

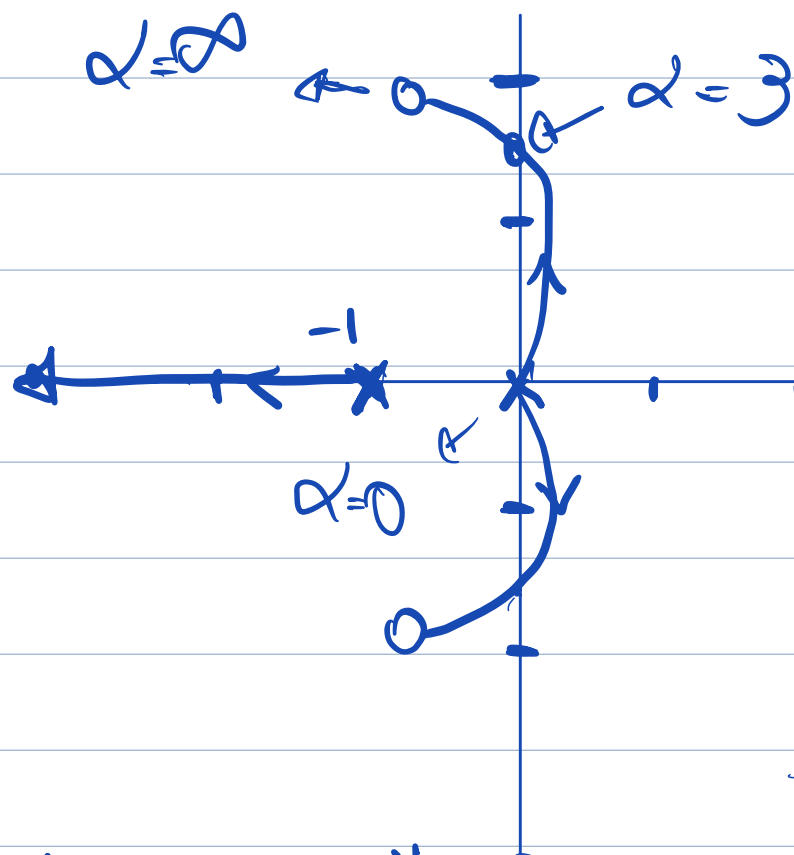
$$-j\omega^3 - \omega^2 + \alpha(-\omega^2 + j\omega + 4) = 0$$

We solve for the real and imaginary part

$$-\omega^2 - \alpha\omega^2 + 4\alpha = 0 \quad -\omega^3 + \alpha\omega = 0$$

$$1) \alpha = 0 \\ \omega = 0$$

$$2) -\omega^2 + \alpha = 0 \Rightarrow \omega^2 = \alpha \\ -\alpha + 4\alpha - \alpha^2 = 0 \\ 3\alpha - \alpha^2 = 0 \\ \alpha = 3 \quad \omega = \pm\sqrt{3}$$



Asymptotically
stable for
 $\alpha > 3$

b) Robustness criteria

$$|G_c(j\omega)| < \frac{1}{|\Delta G(j\omega)|}$$

$\forall \omega$ (1)

$$G^0(s) = G(s) (1 + \Delta G(s))$$

} Real behavior
system

$$G^o(s) = G(s) \frac{\alpha}{(s+\alpha)} = G(s) \left(1 + \frac{\alpha}{s+\alpha} - 1 \right)$$

$$G^o(s) = G(s) \left(1 + \underbrace{\frac{-s}{s+\alpha}}_{\Delta G(s)} \right)$$

$\Delta G(s)!!$

Using (1)

$$\frac{1}{|\Delta G(j\omega)|} = \left| \frac{j\omega + \alpha}{j\omega} \right| \Rightarrow \frac{1}{|\Delta G(j\omega)|} = \frac{\sqrt{\omega^2 + \alpha^2}}{\omega}$$

We want to evaluate sensitivity in a critical ω ...
one that we know has high magnitude that
can affect the robustness criteria.!!

we look at the peak in the bode

$|G_c(j\omega)| \Rightarrow$ max value is at $\omega=2$ with magnitude 2.

$$\text{Then } |2| < \frac{\sqrt{4 + \alpha^2}}{2} \Rightarrow 16 < 4 + \alpha^2$$
$$12 < \alpha^2$$
$$\alpha > \sqrt{12}$$

So for the criterion condition to be fulfilled
 $\alpha > \sqrt{12} \approx 3.46$

- c) a) $\alpha \geq 3$
b) $\alpha > 3.46$

we need $\alpha > 3.46$ to the system to be stable and robust against noise.

Controllability

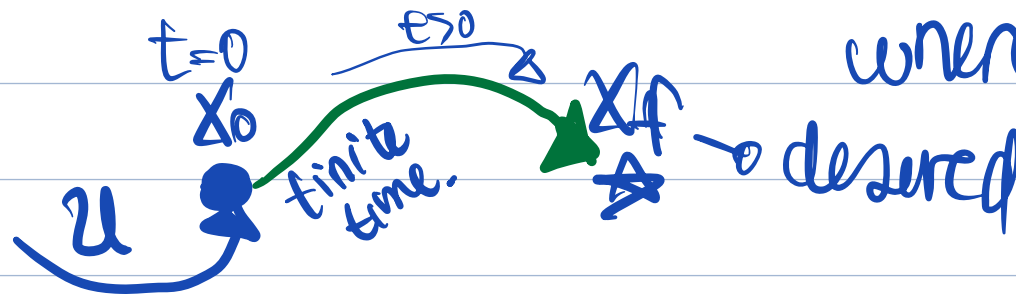
If we have an initial state $x(0)$, the system is controllable if there exists an input sequence that takes $x(0)$ to a desired $x(1)$

$\exists u \in U$ s.t. applying $u(t)$ to the system
Control space
(A, B) we can take x_0 to x_f in a finite time

How to check?

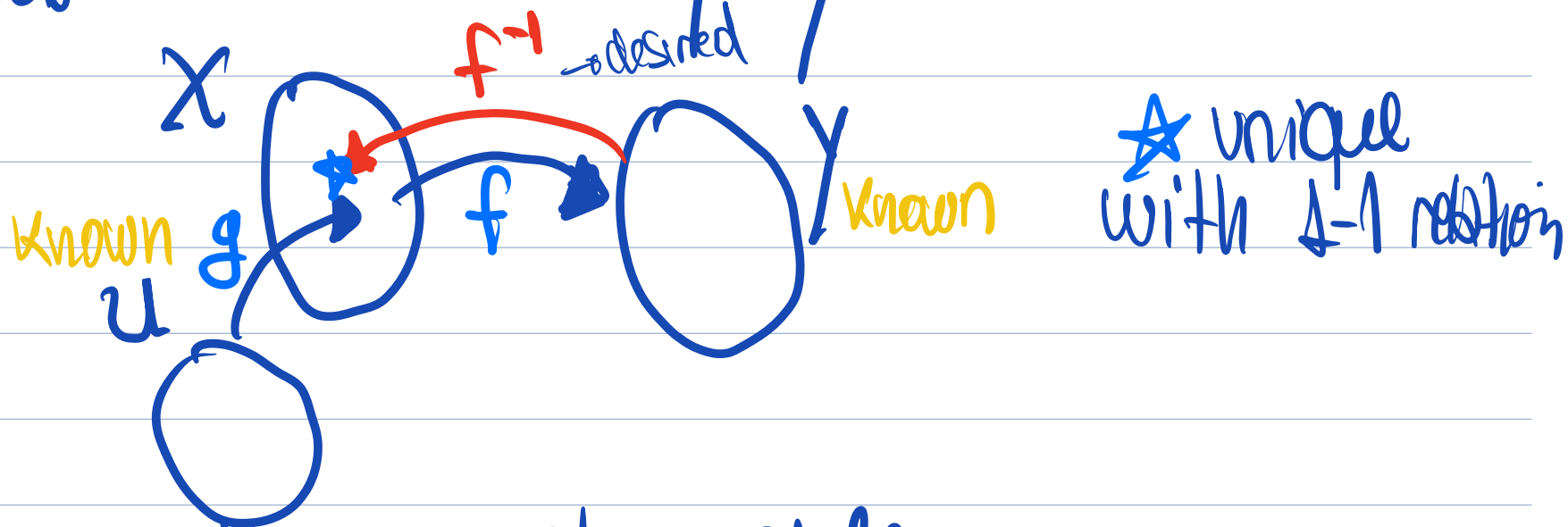
$$C = [A \quad AB \quad A^2B \quad \dots \quad A^{n-1}B] \Rightarrow \text{full rank} = n$$

where A $n \times n$



Observability

Given (u, y) we can distinguish the initial state x_0 which is uniquely



when the system is observable

$$\mathcal{O} = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix} \Rightarrow \text{Full Rank}$$

Transfer function to state space form

$$G(s) = \frac{b_1 \cdot s^{n-1} + \dots + b_{n-1} s + b_n}{s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n}$$

State space into the controllable canonical form

$$\dot{x} = \begin{bmatrix} -a_1 & -a_2 & \dots & -a_{n-1} & -a_n \\ 1 & 0 & & & \\ 0 & 1 & & & \\ \vdots & & \ddots & & \\ 0 & & & 1 & 0 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix} u$$

$$y = [b_1 \ b_2 \ \dots \ b_n] x$$

Observable canonical form

$$\dot{x} = \begin{bmatrix} -a_1 & 1 & 0 & \dots & 0 \\ -a_2 & 0 & 1 & & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ -a_{n-1} & \vdots & \vdots & & 1 \\ -a_n & \vdots & \vdots & & 0 \end{bmatrix} x + \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_{n-1} \\ b_n \end{bmatrix} u$$

$$y = [1 \ 0 \ \dots \ 0]$$

Linearization

Sometimes system are dependant of time (Non linear in time)

$$\dot{x}(t) = f(x, u)$$

$$y(t) = h(x, u)$$

But we want

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) + Du(t)\end{aligned}$$

Then we make this equation linear in an equilibrium point (x_g, u_g) and we rewrite this as,

$$\Delta \dot{x}(t) = \frac{df}{dx}(x_g, u_g) \cdot \Delta x + \frac{df}{du}(x_g, u_g) \Delta u$$

$$\Delta \chi = \frac{dh(x_q, u_q)}{dx} \Delta x + \frac{dh(x_q, u_q)}{du} \Delta u$$

Since we are at an equilibrium point

$$\dot{x}_q = \ddot{x}_q = \dot{u} = \ddot{u} = 0 //$$

8.2 state variables

$$\begin{aligned} x_1 &= \theta & \Rightarrow \dot{x}_1 &= x_2 \\ x_2 &= \dot{\theta} & \dot{x}_2 &= ? \text{ (from equation)} \end{aligned}$$

$\Rightarrow$ Let's write the equation in terms of the state space variables

$$\Rightarrow \ddot{\theta} + \frac{g}{l} \sin \theta + \frac{\ddot{z}}{l} \cos \theta = 0$$

$$\dot{x}_2 + \omega_0^2 \sin x_1 + u(t) \cdot \omega_2 x_1 = 0 \quad (1)$$

The equation is ^{NOT} linear!! we can't represent this using

$$\begin{aligned} \dot{x}(t) &= A x(t) + B u(t) \\ y(t) &= C x(t) + D u(t) \end{aligned} \quad \left. \vphantom{\begin{aligned} \dot{x}(t) &= A x(t) + B u(t) \\ y(t) &= C x(t) + D u(t) \end{aligned}} \right\} \begin{array}{l} \text{STATE SPACE} \\ \text{REPRESENTATION} \end{array}$$

Then we linearize for $x_{1q} = \pi$ $x_{2q} = 0$, $u_q = 0$

We take the derivatives for each term in (1)

$$\Rightarrow \Delta \dot{x}_2 + \omega_0^2 \cdot \cos(x_{1q}) \cdot \Delta x_1 + \Delta u(t) \cos x_{1q} \\ + \underbrace{u_q}_{0} - (\sin(x_{1q})) \Delta x_1 = 0$$

$$\Delta \dot{x}_2 - \omega_0^2 \Delta x_1 - \Delta u(t) = 0$$

$$\Rightarrow \Delta \dot{x}_2(t) = \omega_0^2 \Delta x_1(t) + \Delta u(t)$$

$$\Delta \dot{x}_1 = \Delta x_2$$

$$\begin{bmatrix} \Delta \dot{x}_1(t) \\ \Delta \dot{x}_2(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \omega_0^2 & 0 \end{bmatrix} \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \Delta u$$

$$\Delta y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \end{bmatrix}$$

8.3

We look where there are **derivatives**
(we search for $\frac{1}{s}$ or s)

$\dot{z}$, $\dot{\theta}$, $\dot{\gamma}$ $\Rightarrow$ These are my state variables

$$\begin{aligned} x_1 &= z \\ x_2 &= \theta \\ x_3 &= \gamma \end{aligned}$$

Inputs : M_0 & i
Output : $y(t)$

We write the equations

$$\mathcal{L}^{-1} \chi_1 = (i k_1 - \chi_2 \cdot k_2) \frac{1}{s}$$

- $\dot{\chi}_1(t) = -k_2 \cdot \chi_2(t) + k_1 \cdot i$

$$\mathcal{L}^{-1} (-\chi_3 + \chi_1) \frac{1}{s} = \chi_2$$

- $\dot{\chi}_2(t) = \chi_1(t) - \chi_3(t)$

$$\mathcal{L}^{-1} (\chi_2 \cdot k_2 + M l) \frac{1}{s} = \chi_3$$

- $\dot{\chi}_3(t) = k_2 \cdot \chi_2(t) + M l$

$$\begin{matrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{matrix} = \begin{bmatrix} 0 & -k_2 & 0 \\ 1 & 0 & -1 \\ 0 & k_2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} k_1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} i \\ u \end{bmatrix}$$

$$y(t) = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

8.4a.

$$\frac{d^3 y}{dt^3} + 6 \frac{d^2 y}{dt^2} + 11 \frac{dy}{dt} + 6y = 6u \quad (1)$$

when we have multiple derivatives over the output we do

$$x_1 = y$$

$$x_2 = \dot{y}$$

$$x_3 = \ddot{y}$$

$$\dot{x}_1 = \dot{y} = x_2$$

$$\dot{x}_2 = \ddot{y} = x_3$$

$$x_3 = \dddot{y} = \text{which equation?}$$

(1) !!!

$$\dot{x}_3(t) + 6x_3(t) + 11x_2(t) + 6x_1(t) = 6u$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y(t) = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

If we use the canonical forms
we need to take the laplace

transform of (1)

$$s^3 Y(s) + 6s^2 Y(s) + 11s Y(s) + 6Y = 6u$$

$$\frac{Y(s)}{U(s)} = \frac{6}{s^3 + 6s^2 + 11s + 6}$$

$$G(s) = \frac{b_1 \cdot s^{n-1} + \dots + b_{n-1} s + b_n}{s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n}$$

$$\{ b_1 = 6 \quad ; \quad a_1 = 6 \quad a_2 = 11 \quad a_3 = 6 \}$$

$$b_2 = 0$$

$$b_3 = 0$$

Controllable canonical form

$$\dot{x} = \begin{bmatrix} -6 & 1 & 6 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} x + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} u$$

$G(s) = \frac{b_1 s^{n-1} + \dots + b_{n-1} s + b_n}{s^n + a_1 s^{n-1} + \dots + a_{n-1} s + a_n}$

$$y = [6 \ 0 \ 0] x$$

8.4b

What if we have $\ddot{u}$?

Let's start by making $x_1(t) = y(t)$
and we move all the derivatives to the left

$$\frac{d^3}{dt^3} x_1(t) + \frac{d^2}{dt^2} x_1(t) + 5 \frac{d}{dt} x_1(t) - 4 \frac{d^2}{dt^2} u(t)$$

$$- \frac{d}{dt} u(t) = -3 x_1(t) + 2 u(t)$$

$$\frac{d}{dt} \left[\frac{d^2}{dt^2} x_1 + \frac{d}{dt} x_1 + 5 x_1 - 4 \frac{d}{dt} u(t) - u(t) \right] = 0$$

New state variable 4)

$$x_2 = \frac{d^2}{dt^2} x_1 + \frac{d}{dt} x_1 + 5x_1 - \frac{4}{dt} u(t) - u(t)$$

$$\dot{x}_2(t) = -3x_1(t) + 2u(t)$$

$$\frac{d}{dt} \left[\frac{d}{dt} x_1 + x_1 - 4u(t) \right] = x_2 - 5x_1 + u(t)$$

$$x_3 = \dot{x}_1(t) + x_1 - 4u(t)$$

$$\dot{x}_3 = x_2 - 5x_1 + u(t)$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -1 & 0 & 1 \\ -3 & 0 & 0 \\ -5 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix} u(t)$$

$$y = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} x$$

Easy way (canonical forms)

$$\frac{Y(s)}{U(s)} = \frac{4s^2 + s + 2}{s^3 + s^2 + 5s + 3}$$

$$b_1=4 \quad b_2=1 \quad b_3=2$$

$$a_1=1 \quad a_2=5 \quad a_3=3$$

8.4c

$$G(s) = \frac{2s+3}{s^2+5s+6}$$

$$a_1 = 5 \quad a_2 = 6 \quad b_1 = 2 \quad b_2 = 3$$

① Controllable canonical form

$$\dot{x} = \begin{bmatrix} -a_1 & -a_2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u$$

$$y = [b_1 \quad b_2] x$$

$$\dot{x} = \begin{bmatrix} -5 & -6 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u \quad y = [2 \quad 3] \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

② Observable canonical form

$$\dot{x} = \begin{bmatrix} -a_1 & 1 \\ -a_2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} u \quad y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\dot{x} = \begin{bmatrix} -5 & 1 \\ -6 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 2 \\ 3 \end{bmatrix} u \quad y = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

③

$$G(s) = \frac{Y(s)}{U(s)} = \frac{2s+3}{s^2+s+6}$$

$$s^2 y + s \dot{y} + 6y = 2\dot{u} + 3u \quad ??$$

What is that??
How can we treat this

$$G(s) = \frac{A}{(s+3)} + \frac{B}{(s+2)}$$

$$2s+3 = A(s+2) + B(s+3)$$

$$s = -2$$

$$-1 = B(1) \Rightarrow B = -1$$

$$s = -3$$

$$-3 = A(-1) \Rightarrow A = 3$$

$$G(s) = \frac{3}{(s+3)} - \frac{1}{(s+2)}$$

$$Y(s) = \frac{3}{(s+3)} U(s) - \frac{1}{(s+2)} U(s)$$

$$x_1 = \frac{3}{(s+3)} u(s)$$

$$x_2 = \frac{1}{(s+2)} u(s)$$

$$\dot{x}_1 + 3x_1 = 3u(s)$$

$$\dot{x}_2 + 2x_2 = u(s)$$

$$y(t) = x_1 - x_2$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -3 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 3 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} //$$

state space description

$M_l, i \Rightarrow$ inputs
 $y \Rightarrow$ output

we determine the states based on the integrators.

In this case

$$\begin{aligned}x_1(t) &= y_1(t) & x_2(t) &= \theta(t) \\x_3(t) &= z(t).\end{aligned}$$

Then we write the equation for each state

$$\dot{x}_1(t) = \frac{1}{s} [M_l + \theta K_2] = \frac{1}{s} [M_l + x_2(t) \cdot K_2] \quad (1)$$

$$\dot{x}_2(t) = \frac{1}{s} [z - y] = \frac{1}{s} [x_3(t) - x_1(t)] \quad (2)$$

$$\dot{x}_3(t) = \frac{1}{s} [i K_i - \theta K_2] = \frac{1}{s} [i \cdot K_i - x_2(t) K_2] \quad (3)$$

we multiply each by s and take the Laplace inverse

$$\dot{x}_1(t) = k_2 x_2(t) + M_0$$

$$\dot{x}_2(t) = -x_1(t) + x_3(t)$$

$$\dot{x}_3(t) = -k_2 x_2(t) + i k_i$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} 0 & k_2 & 0 \\ -1 & 0 & 1 \\ 0 & -k_2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \underbrace{\begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & k_i \end{bmatrix}}_{\text{input matrix}} \begin{bmatrix} M_0(t) \\ i(t) \end{bmatrix}$$

$$y(t) = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} M_0(t) + \begin{pmatrix} 0 \\ 0 \\ k_i \end{pmatrix} i(t)$$

can be also
written like this!

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$$\dot{x}_1(t) = \sin(x_2(t) + u(t))$$

$$\dot{x}_2(t) = \sin(u(t))$$

$$y(t) = x_1(t) - u(t)$$

stationary : $\dot{x}_1 = \dot{x}_2 = 0$

$$\sin(x_{2ss} + u_{ss}) = 0 \quad \sin(u_{ss}) = 0$$

$$y_{ss} = x_{1ss} - u_{ss}$$

$$u_{ss} = \pi$$

we can pick $x_2(t) = k\pi$
 $k = \{0, 1, 2, 3\}$

For $K=0$ $x_2=0$
 x_1 is free then $x_1=0$

We linearize

$$\Delta \dot{x}_1(t) = \cos(x_{2ss} + u_{ss}) \Delta x_2(t) + \cos(x_{2ss} + u_{ss}) \Delta u(t)$$

$$\Delta \dot{x}_2(t) = \cos(u_{ss}) \Delta u(t)$$

$$\Delta y(s) = \Delta x_1 \cdot u_{ss} + \Delta u \cdot x_{1ss}$$

$$\rightarrow \Delta \dot{x}_1(t) = -\Delta x_2(t) - \Delta u(t)$$

$$\rightarrow \Delta \dot{x}_2(t) = -\Delta u(t)$$

$$\rightarrow \Delta y(s) = \pi \Delta x_1$$